

Customer Shopping Behavior Analysis

1. Project Overview

This project examines customer shopping behavior using transactional data from 3,900 purchases across different product categories. The primary objective is to identify patterns in customer spending, analyze customer segments, understand product preferences, and evaluate subscription behavior. The insights gained from this analysis can support informed strategic business decisions and improve overall business performance.

2. Dataset Summary

- Rows: 3900
- Columns: 18
- Key features:
 - Customer Demographic: Information related to customers, including their age, gender, location and subscription status.
 - Purchase Information: Details about transactions, such as the purchase item, product category and purchase amount and season of purchase, size and color preferences.
 - Shopping Behavior: Data reflecting customer purchasing habits, including discount usage, promotional code application, previous purchase history, purchase frequency, review ratings and preferred shipping methods.
- Missing Data: 37 values in review Rating column.

3. Exploratory Data Analysis using python.

3.1 Dataset Overview:

The customer shopping behavior dataset contains 3,900 transaction records and includes variables related to customer demographics, purchasing information, and shopping behavior. The dataset provides valuable insights into customer preferences, spending patterns, and subscription behavior, which were analyzed using Python-based exploratory data analysis techniques.

3.2 Data Cleaning:

deduplicated() duplicate records were identified and removed. The dataset was checked for data type consistency and standardized for further analysis.

```
df.duplicated().sum()
```

```
np.int64(0)
```

3.3 Column Standardization:

Renamed columns for better readability and documentation.

```
df.columns=df.columns.str.lower()  
df.columns=df.columns.str.replace(' ','_')  
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount_usd', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'promo_code_used', 'previous_purchases',  
      'payment_method', 'frequency_of_purchases'],  
      dtype='object')
```

3.3 Missing Value Analysis

`isnull().sum()` missing values were examined using the function.

```
df.isnull().sum()
```

```
Customer ID      0  
Age              0  
Gender           0  
Item Purchased   0  
Category         0  
Purchase Amount (USD)  0  
Location         0  
Size            0  
Color           0  
Season          0  
Review Rating    37  
Subscription Status  0  
Shipping Type    0  
Discount Applied  0  
Promo Code Used  0  
Previous Purchases  0  
Payment Method   0  
Frequency of Purchases  0  
dtype: int64
```

3.4 Data Consistency Check:

Verified it `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.

```
df[['discount_applied', 'promo_code_used']].head(10)
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

3.5 Database Integration:

Connected python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

```
import pandas as pd
from sqlalchemy import create_engine

username = "root"
password = ""
host = "localhost"
port = "3306"
database = "customer_behaviour"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}"
)

print("Connected successfully!")
```

Connected successfully!

```
df.to_sql(
    name='customers',
    con=engine,
    if_exists='replace',
    index=False
)

print("Data uploaded successfully!")
```

Data uploaded successfully!

4. Data Analysis using SQL

1. What is the total revenue generated by male vs. female customers?

Result Grid			Filter
	gender	revenue	
▶	Female	75191	
	Male	157890	

2. Which customer used a discount but still spent more than the average purchase amount?

Result Grid			Filter Rows:
	customer_id	purchase_amount	
▶	2	64	
	3	73	
	4	90	
	7	85	
	9	97	
	12	68	
	13	72	
	14	81	
	15	84	

3. Which are the top five products with highest average review rating?

	item_purchased	Average Product Rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.80
	Skirt	3.78

4. Compare the average purchase amount between standard and express shipping?

	shipping_type	ROUND(AVG (purchase_amount),2)
▶	Express	60.48
	Standard	58.46

5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers?

Result Grid					Filter Rows:	Export:	Wrap
	subscription_status	total_customers	avg_spend	total_revenue			
▶	Yes	1053	59.49	62645.00			
	No	2847	59.87	170436.00			

6. Which 5 products have the highest percentage of purchase with discounts applied?

Result Grid			Filter Rows:
	item_purchased	discount_rate	
▶	Hat	50.00	
	Sneakers	49.66	
	Coat	49.07	
	Sweater	48.17	
	Pants	47.37	

7. Segment customers into new, returning and loyal based on their total number of previous purchase, and show the count of each segment?

	customer_segment	number of customers
►	loyal	3116
	new	83
	Returning	701

8. What are the top 3 most purchase products within each category?

	item_rank	category	item_purchased	total_orders
►	3	Accessories	Belt	161
	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171

Result 9 x

9. Are customer who are repeat buyers (more than 5 previous purchases) also likely to subscribers?

	subscription_status	repeat_buyers
►	No	2518
	Yes	958

10. What is the revenue contribution of each age group?

	age_group	total_revenue
►	Young Adult	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763

5. Dashboard in Power BI



6. Business Recommendation:

Based on the analysis of customer shopping behavior, several strategic recommendations can be made to improve customer satisfaction, increase revenue, and enhance business performance:

- Focus marketing efforts on high-value and subscribed customers.
- Increase sales through targeted discounts and promotional campaigns.
- Prioritize inventory management for top-selling products and categories.
- Implement customer segmentation strategies for personalized marketing.
- Strengthen subscription and loyalty programs to improve customer retention.
- Optimize shipping services according to customer preferences.
- Monitor customer reviews and ratings to enhance product quality and customer satisfaction.
- Utilize seasonal purchasing trends to plan inventory and promotional activities effectively.